

Saa'im Abdullah

Software Engineer | Lahore, Pakistan | +92 324 9923289

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Software engineer building customer-facing applications, distributed services, and data pipelines. Experience spans product delivery, database performance, asynchronous processing, and cloud deployment.

Experience

Fitter Health

Software Engineer, Contract

May 2026 – Jul 2026

Remote / Spain-based team

- Delivered membership credits, coach booking, commerce, and CRM workflows across a **106-table PostgreSQL schema**. Integrated a Next.js frontend with Django APIs and handed over documented services with **135 passing tests**.
- Fixed silent notification loss through SNS-to-SQS buffering, retries, dead-letter handling, and queue-depth alarms. Automated renewal and re-engagement email with scheduled Lambda/SES workflows, replacing manual follow-up.
- Deployed containerized services on ECS with RDS and S3, configuring IAM and VPC access. Implemented Celery/Redis jobs and Odoo invoice reconciliation to synchronize payment status with application records.
- Shipped ETL into a PostgreSQL star-schema warehouse supporting **three use cases**, ML features, product analytics, and aggregated partner reporting.

Expertflow – Cisco UCCE Solution Partner

Software Engineer

Jun 2024 – Feb 2026

Lahore, Pakistan

- Reduced admin panel response time from **130 ms to 30 ms, a 77% improvement**, through Redis caching and database index tuning.
- Built APIs serving **1,000–3,000 daily users**, deployed Docker services on EC2 behind a load balancer, and applied rate limiting and time-bound access controls. Configured EKS autoscaling based on request throughput.
- Developed a WebRTC SDK for customer voice, chat, and video interactions with frontend and third-party API integrations. Packaged, versioned, and published [customer-js-sdk-expertflow](#) npm modules.
- Enhanced Angular admin interfaces and built survey forms rendered dynamically in vanilla JavaScript, supporting multiple question types and backend integration. Added chatbot responses, SQL analytics, and customer dashboards.
- Automated batch and streaming ingestion with Lambda, Airflow, PySpark, EMR, Glue, and S3, transforming and cataloging raw inputs as Parquet datasets for BI reporting and ML workflows.

Projects

Multilingual, Multi-Tenant RAG Engine | Freelance | [Code](#) | [Article](#)

Built admin ingestion for CSV, PDF, and DOCX documents and customer-facing Q&A APIs using Django, pgvector, multilingual embeddings, and Gemini. Enforced authenticated tenant-scoped retrieval, quotas, and **six prompt templates** across isolated knowledge bases.

Real-Time E-Commerce ETL Pipeline | Freelance | [Code](#) | [Article](#)

Built Kafka/PySpark streaming ingestion with checkpoints, date-partitioned Parquet, validation, and deduplication. Orchestrated hourly Airflow transforms through bronze, silver, and gold layers into a **four-table sales star schema**.

Movie Recommender System | [Code](#)

Combined MovieLens content similarity, co-occurrence, and popularity for cold-start recommendations. Separated training, evaluation, and serving, with FastAPI recommendation, health, and model-reload APIs, API-key protection, Redis caching, Docker, and request tracing.

Autoencoder for Edge Sensor Data | [Code](#)

Deployed a TensorFlow Lite Micro encoder on ESP32 to compress sensor readings and send latent vectors to a Flask decoder on EC2 with S3 integration. Evaluated model-size, inference-speed, and reconstruction trade-offs.

Additional work. Cloud intrusion detection and secure IoT communication, hackathon EC2 deployment and backend integration, Bee Simulator Unity prototype, [Python algorithms and complexity notes](#), ETL and RAG technical articles.

Technical Skills

Languages. Python, Java, JavaScript, TypeScript, SQL, C++, C#

Services and applications. gRPC, Protocol Buffers, REST APIs, WebRTC, Django/DRF, FastAPI, Flask, Node.js, Express, React, Next.js, Angular

Data and infrastructure. PostgreSQL, Redis, MongoDB, MySQL, DynamoDB, Kafka, PySpark, Airflow, pgvector, TensorFlow, scikit-learn, AWS, Docker, Kubernetes, CI/CD, Git, Linux, Postman, Jira

Education and Courses

COMSATS University Islamabad, Lahore Campus

2021 – 2025

BS in Computer Engineering. Coursework in Data Structures and Algorithms, Databases, Operating Systems, Networks, Software Engineering, Computer Architecture, and Machine Learning.

Courses. [AWS Cloud Technical Essentials](#), [Introduction to Data Engineering](#), [Getting Started with Git and GitHub](#), [Claude Code 101](#).